LAB 7

6. $X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$ $W_Q = W_K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$

$Q = X \cdot W_Q = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ $K = K \cdot W_K = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ $V = X W_V = \begin{bmatrix} 2 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$

$S = \frac{QK^T}{\sqrt{2}} = \begin{bmatrix} 2,8284 & 0 & 1,4142 \\ 0 & 2,8284 & 1,4142 \\ 1,4142 & 1,4142 & 1,4142 \end{bmatrix}$ $K^T = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$

$A_{ij} = \frac{e^{S_{ij}}}{\sum_j e^{S_{ij}}} \Rightarrow A \approx \begin{bmatrix} 0,7679 & 0,0454 & 0,1867 \\ 0,0454 & 0,7679 & 0,1867 \\ 0,3333 & 0,3333 & 0,3333 \end{bmatrix}$

$Z = AV \approx \begin{bmatrix} 1,7225 & 1 \\ 0,2775 & 1 \\ 1 & 1 \end{bmatrix}$

7. $X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$ $W_{Q_1} = W_{K_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_{V_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$

Tương tự như trên $head_1 = A_1 V_1 \approx \begin{bmatrix} 1,7225 & 1 \\ 0,2775 & 1 \\ 1 & 1 \end{bmatrix}$

$W_{Q_2} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ $W_{K_2} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ $W_{V_2} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$

$Q_2 = XW_{Q_2} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ $K_2 = XW_{K_2} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \end{bmatrix}$ $V_2 = XW_{V_2} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$

$S_2 = \frac{Q_2 K_2^T}{\sqrt{2}} = \begin{bmatrix} 3 & 4 & 4 \\ 2 & 3 & 3 \\ 3 & 4 & 4 \end{bmatrix}$ $S_2 = \frac{Q_2 K_2^T}{\sqrt{2}} = \begin{bmatrix} 2,1213 & 2,8284 & 2,8284 \\ 1,4142 & 2,1213 & 2,1213 \\ 2,1213 & 2,8284 & 2,8284 \end{bmatrix}$

$A_2 = \text{softmax}(S_2) \approx \begin{bmatrix} 0,1978 & 0,4011 & 0,4011 \\ 0,1978 & 0,4011 & 0,4011 \\ 0,1978 & 0,4011 & 0,4011 \end{bmatrix}$

$head_2 \approx \begin{bmatrix} 1,4011 & 1,1978 \\ 1,4011 & 1,1978 \\ 1,4011 & 1,1978 \end{bmatrix}$

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$C = \text{concat}(\text{head}_1, \text{head}_2)$

$$\approx \begin{bmatrix} 1,7225 & 1 & 1,4011 & 1,1978 \\ 0,2775 & 1 & 1,4011 & 1,1978 \\ 1 & 1 & 1,4011 & 1,1978 \end{bmatrix} \qquad W^o = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\text{Multihead} \approx \begin{bmatrix} 3,1256 & 2,1978 & 2,4011 & 2,9203 \\ 1,6786 & 2,1978 & 2,4011 & 1,4753 \\ 2,4011 & 2,1978 & 2,4011 & 2,1978 \end{bmatrix}$$

Cần concatenation trước khi áp dụng W^o vì concatenation giúp đầu ra của các attention heads thành một representation chung, giữ lại thông tin từng head. $\xrightarrow{W^o}$ biến đổi thành → dimension mong muốn.

10 pos = 0 → $PE(0) = [0 \quad 1 \quad 0 \quad 1] = [\sin 0 \quad \cos 0 \quad \sin 0 \quad \cos 0]$

pos = 1 → $PE(1) = [\sin 1 \quad \cos 1 \quad \sin 0{,}01 \quad \cos 0{,}01] \approx$

$= [0{,}8415 \quad 0{,}5403 \quad 0{,}0099 \quad 0{,}9999]$

pos 2 → $PE(2) \approx [\sin 2 \quad \cos 2 \quad \sin 0{,}02 \quad \cos 0{,}02]$

$\approx [0{,}9093 \quad -0{,}4161 \quad 0{,}0199 \quad 0{,}9998]$

14 $W_1 x = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$h = \text{ReLU}(z) = \begin{bmatrix} \max(0,1) \\ \max(0,1) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$W_2 = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow y = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

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